

MobileVidFactory: Automatic Diffusion-Based Social Media Video Generation for Mobile Devices from Text

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ABSTRACT

Videos for mobile devices become the most popular access to share and acquire information recently. For the convenience of users' creation, in this paper, we present a system, namely **MobileVidFactory**, to automatically generate vertical mobile videos where users only need to give simple texts mainly. Our system consists of two parts: basic and customized generation. In the basic generation, we utilize the pretrained image diffusion model, and adapt it to a high-quality open-domain vertical video generator. As for the audio, by retrieving from our big database, our system matches a suitable background sound for the video. Additionally to produce customized content, our system allows users to add specified screen texts for enriching visual expression, and specify texts for automatic reading with optional voices as they like.

CCS CONCEPTS

• Information systems → Multimedia content creation; • Computing methodologies → Computer vision.

KEYWORDS

Vertical Video Generation, Diffusion Model, Mobile Video

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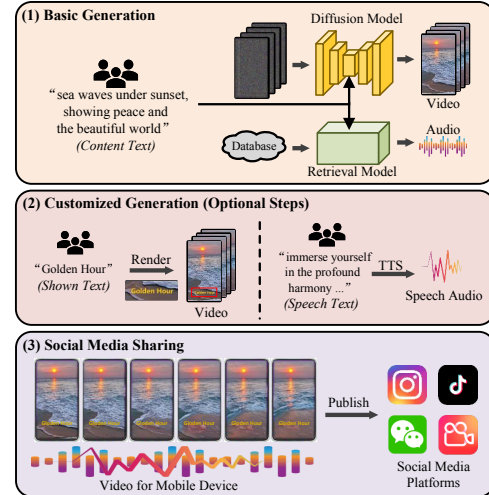


Figure 1: Illustration of our system. Given an input text, our model generates the basic visual content and retrieves the audio part for the video. Then, users can further add visible texts to the decorate visual part or human speech to enrich the expression with texts as well.

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1 INTRODUCTION

The proliferation of mobile devices, particularly smartphones, and the explosive growth of social media have revolutionized video consumption. Vertical videos have gained prominence among mobile users, driving our investigation into the challenges, opportunities, and implications for creators and consumers. This study explores the transformative potential of this evolving visual storytelling form. However, crafting such videos demands professional skills

across shooting and editing. In this paper, we present a novel system **MobileVidFactory** to elegantly help users create their own videos for mobile devices. In summary, our system is leading in the following aspects:

- 1) Our system is the first automatic mobile-device video generation framework to the best of our knowledge, which can be used to generate high-quality, sounding, and vertical videos with simple text from users.
- 2) Our system contains both deep networks to generate the basic visual and auditory content, and fixed modules to add customized content for users, where all components are controlled by text mainly. This makes our system meet different needs of users as much as possible and operate easily.
- 3) Our system is able to learn to synthesize open-domain high-quality vertical videos after finetuning with a few vertical video data, as we design an efficient training strategy to learn motion information from large-scale horizontal video datasets and migrate it losslessly to vertical video.

2 SYSTEM

As shown in Fig.1, our system contains two generation processes: basic generation, which produces the elemental video and audio, and customized generation, which offers the option to add screen text and dubbing.

2.1 Basic Generation

In the basic generation, not like prior works [6, 10], we independently produce the visual and auditory content. The video is synthesized using a diffusion model, and the audio is obtained through a retrieval model. To generate high-quality vertical videos, we pre-train our model on a large-scale horizontal video dataset [1, 15], and adapt it to the vertical generation using a few vertical data. Specifically, we firstly extend a pretrained image diffusion model [8, 12] to fit the spatial distribution (e.g., new aspect ratio and picture style) of the video dataset by adding learnable blocks in each level of the U-Net [9] network, and only train these blocks while fixing all pretrained layers. Then, building upon previous works [2, 11], we add new temporal layers, including temporal 1D residual blocks and transformer blocks [13], and utilize the video dataset to train these additional layers [16]. After pretraining, we fix all layers except for the ones added to adjust the spatial distribution. We then finetune these spatial blocks to adapt to the vertical screens of mobile devices using some collected data. Additionally, we provide a frame interpolation module [4, 5] to achieve smoother motion in certain cases. To match vivid audio to the visual content, we adopt the previous work [3, 7] as our retrieval model. The network employs text and audio encoders to process data and assesses feature distances to locate relevant samples, trained with contrastive ranking loss. In our scenario, we directly use user-provided text to directly retrieve the top three matching audios from the database, enabling users to select preferred audio and seamlessly integrate it into the video.

2.2 Customized Generation

In our system, we design two optional customized functions for users. First, we support adding customized screen text to the video, as the addition of text to short mobile videos can provide context,



Figure 2: Generated video samples of our system. More samples are included in our video.

clarify concepts, and enhance viewer engagement. Text overlays also promote accessibility for individuals with hearing impairments and cater to diverse audiences. With the provided text to be shown, the specified font size and font color to use, and the designated position to place, our system directly overlays the text on top of the video for rendering. Second, to add a personal touch, enhance storytelling, provide explanations, and foster a sense of authenticity, our system allows users to add dubbing to their videos. Using sentences from users, we utilize bark [14] to transform the text into dubbing, known as text-to-speech (TTS). In this process, users can choose the voice they prefer, and various languages, such as English and Chinese, are well supported.

2.3 Demonstration

We demonstrate some generated videos in Fig.2. Focusing on the vertical videos for mobile devices, our system can synthesize frames with abundant details and create compositions that highlight the subject. Additionally, smooth motion can be captured to describe vivid scenes. More samples are included in our video.

3 CONCLUSION

In this paper, we present MobileVidFactory, a system that automatically generates vertical videos for mobile devices using text inputs. The system consists of a visual generator that creates high-quality videos by leveraging a pretrained image diffusion model and a finetuning process. Users can enrich visual expression by adding visible specified texts. The audio generator matches suitable background sounds from a database and provides optional text-to-speech narration.

REFERENCES

- [1] Max Bain, Arsha Nagrani, Gül Varol, and Andrew Zisserman. 2021. Frozen in Time: A Joint Video and Image Encoder for End-to-End Retrieval. In *ICCV*.
- [2] Andreas Blattmann, Robin Rombach, Huan Ling, Tim Dockhorn, Seung Wook Kim, Sanja Fidler, and Karsten Kreis. 2023. Align your Latents: High-Resolution Video Synthesis with Latent Diffusion Models. In *CVPR*.
- [3] A. Sophia Koepke, Andreea-Maria Oncescu, João F. Henriques, Zeynep Akata, and Samuel Albanie. 2022. Audio Retrieval with Natural Language Queries: A Benchmark Study. *IEEE TMM* (2022).
- [4] Zhen Li, Zuo-Liang Zhu, Linghao Han, Qibin Hou, Chun-Le Guo, and Ming-Ming Cheng. 2023. AMT: All-Pairs Multi-Field Transforms for Efficient Frame Interpolation. In *CVPR*.
- [5] Chengxu Liu, Huan Yang, Jianlong Fu, and Xueming Qian. 2022. TTVFI: Learning Trajectory-Aware Transformer for Video Frame Interpolation. *arXiv* (2022).
- [6] Jiawei Liu, Weining Wang, Sihao Chen, Xinxin Zhu, and Jing Liu. 2023. Sounding Video Generator: A Unified Framework for Text-guided Sounding Video Generation. *arXiv* (2023).
- [7] Andreea-Maria Oncescu, A. Sophia Koepke, João F. Henriques, Zeynep Akata, and Samuel Albanie. 2021. Audio Retrieval with Natural Language Queries. In *Interspeech*.
- [8] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *CVPR*.
- [9] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *MICCAI*.
- [10] Ludan Ruan, Yiyang Ma, Huan Yang, Huiguo He, Bei Liu, Jianlong Fu, Nicholas Jing Yuan, Qin Jin, and Baining Guo. 2023. MM-Diffusion: Learning Multi-Modal Diffusion Models for Joint Audio and Video Generation. (2023).
- [11] Uriel Singer, Adam Polyak, Thomas Hayes, Xi Yin, Jie An, Songyang Zhang, Qiyuan Hu, Harry Yang, Oron Ashual, Oran Gafni, et al. 2022. Make-A-Video: Text-to-video generation without text-video data. *arXiv* (2022).
- [12] Stability-AI. 2022. Stable Diffusion. <https://github.com/Stability-AI/StableDiffusion>.
- [13] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is All you Need. In *NeurIPS*.
- [14] Chengyi Wang, Sanyuan Chen, Yu Wu, Ziqiang Zhang, Long Zhou, Shujie Liu, Zhuo Chen, Yanqing Liu, Huaming Wang, Jinyu Li, Lei He, Sheng Zhao, and Furu Wei. 2023. Neural Codec Language Models are Zero-Shot Text to Speech Synthesizers. *arXiv* (2023).
- [15] Wenjing Wang, Huan Yang, Zixi Tuo, Huiguo He, Junchen Zhu, Jianlong Fu, and Jiaying Liu. 2023. VideoFactory: Swap Attention in Spatiotemporal Diffusions for Text-to-Video Generation. *arXiv* (2023).
- [16] Junchen Zhu, Huan Yang, Huiguo He, Wenjing Wang, Zixi Tuo, Wen-Huang Cheng, Lianli Gao, Jingkuan Song, and Jianlong Fu. 2023. MovieFactory: Automatic Movie Creation from Text using Large Generative Models for Language and Images. *arXiv* (2023).